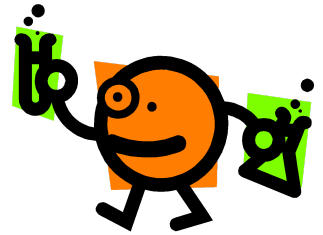


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Grade Level: All are 7th grade



Science and Engineering Fair Project Ideas **"Is My Project Feasible?"**

In addition to the "Project Ideas" you list below, ask yourself...

- 1) Do I have enough time? (Think about the number of weeks you have to collect data.)**
- 2) Can I obtain the necessary materials?**
- 3) Is it safe, and will it be approved by my parents/guardian and teacher?**
- 4) Can it be completed where I live? (For example, we cannot study coral reefs in North Texas. We do not live near them. Also, if you plan on growing any bacteria or mold, you must do this in a science lab under adult supervision. No, Fowler's science lab does not count. This is usually a college or university.)**
- 5) Is this something that is competitive? Is it unique? Can you Google the answer?**
- 6) [Look over this No-No list](#). These are some ideas of things to avoid or tips when coming up with your ideas.**

Project Idea #1 - P.E.N.S.I.L.:

Planetary Expeditionary Navigator and Settlement Infrastructure Launcher

Problem(What are you investigating? Write this in the form of a question) - Can we design a rover system (possibly with multiple programmed swarms of rovers) that is faster, more efficient, and reliable in the factor of planetary colonization/settlement?

Hypothesis(May be written as an IF/THEN statement, or testable statement. This is where you are predicting the outcome of your problem.) or Engineering Goal (If you are doing an engineering project, you won't have a hypothesis. You will have the expected goal of what you are designing.) : We can create a multi-rover system that operates

efficiently, has the ability to assess the planetography of a planetary body, and explore a planet faster than conventional rovers, such as Perseverance.

Independent Variable (This is the tested variable that you are changing.): Type of rover(traditional, e.g. Perseverance v.s. rover system - our design, P.E.N.S.I.L.)

Dependent Variable (This is the data you are gathering. It changes because it is dependent on the independent variable.): speed (use timer and sim), coverage (use rover's GPS updates, incl. in code)

What materials do you think you will need? What tools/instruments will you use for measurement? (This list can be changed once your idea is approved, but you need to have a basic idea of how to test your idea before we can approve it.)

-
- Sensors
 - Coding terminal (for algorithm and GPS Mapping)
 - Nitro buggy car updated to match scientific experiment or batteries
 - Computer chip with movement algorithm
 - Metal encasement
 - Timer
-

Teacher Suggestions (yes, no, improvements, changes, etc.)

Project Idea #2 NOVA: Novel Orbital Velocity Accelerators

Problem(What are you investigating? Write this in the form of a question) -

Is it possible to power a spacecraft through light propulsion to gain near-light speeds?

Hypothesis(May be written as an IF/THEN statement, or testable statement. This is where you are predicting the outcome to your problem.) or Engineering Goal (If you are doing an engineering project, you won't have a hypothesis. You will have the expected goal of what you are designing.): We will attempt to create a spacecraft that is powered through light propulsion for thrust. This design attempts to reach near-light speeds using this technique, and correct the mistakes created by the theoretical version of Laser-Power Ramjet.

Independent Variable (This is the tested variable that you are changing.): Type of spacecraft (without light propulsion[e.g. the fastest spacecraft built that depends solely on its engine: New Horizons] versus NOVA)

Dependent Variable (This is the data you are gathering. It changes because it is dependent on the independent variable.): speed

What materials do you think you will need? What tools/instruments will you use for measurement? (This list can be changed once your idea is approved, but you need to have a basic idea of how to test your idea before we can approve it.)

-
- D-T Combustion Chamber
 - Fusion Reactor
 - Laser beam
 - Pumps
 - Batteries
 - Photons
 - Cryocooler
 - Gimbal
 - Wires
 - Generator

- Photons
-

Teacher Suggestions (yes, no, improvements, changes, etc.)

Project Idea #3 STELLAR-Suborbital Tropism Experiment with Light and Loading for Aerospace Research

Problem(What are you investigating? Write this in the form of a question) - How does the direction of the roots of a plant change in microgravity?

Hypothesis(May be written as an IF/THEN statement, or testable statement. This is where you are predicting the outcome to your problem.) or Engineering Goal (If you are doing an engineering project, you won't have a hypothesis. You will have the expected goal of what you are designing.): If a plant is grown in microgravity with all of its necessities, then the roots will grow in different directions.

Independent Variable (This is the tested variable that you are changing.): Gravity rate (Earth's gravity or microgravity)

Dependent Variable (This is the data you are gathering. It changes because it is dependent on the independent variable.): Roots' direction

What materials do you think you will need? What tools/instruments will you use for measurement? (This list can be changed once your idea is approved, but you need to have a basic idea of how to test your idea before we can approve it.)

-
- Plant seeds
 - Petri Dishes

- Tweezers
 - Growth Chamber
 - Camera
 - LED System
 - Power Source
 - Watering System
 - Arduino Clinostat
-

Teacher Suggestions (yes, no, improvements, changes, etc.)
